



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B. Tech (CBCS) Degree Examinations

Summer Semester July 2024

EC2105-1 – NETWORK THEORY AND CONTROL SYSTEMS

(EC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Potential difference in electrical terminology is known as?**
A) Voltage
B) Resistance
C) Current
D) Conductance
- How many types of dependent or controlled sources are there?**
A) 1
B) 2
C) 3
D) 4
- If the resistances 3Ω , 5Ω , 7Ω , 9Ω are in series, then their equivalent resistance(Ω) is?**
A) 9
B) 20
C) 24
D) 32
- A semiconductor diode is an _____ element.**
A) Bilateral
B) Unilateral
C) Active
D) Passive
- In Superposition theorem, while considering a source, all other current sources are?**
A) short circuited
B) change its position
C) open circuited
D) removed from the circuit
- Thevenin's voltage is equal to the _____ voltage across the _____ terminals.**
A) short circuit, input
B) open circuit, input
C) short circuit, output
D) open circuit, output
- If the source Z_s is complex, then the condition for the maximum power to be transferred is?**
A) $Z_L = Z_s$
B) $Z_L = -Z_s$
C) $Z_L = Z_s^*$
D) $Z_L = -Z_s^*$
- Superposition principle states that at a time _____ source(S) acts.**
A) All the given sources
B) Only current sources
C) Only voltage sources
D) Only one source
- In two-port networks the parameter h_{22} is called _____**
A) Open circuit output admittance
B) Open circuit reverse voltage gain
C) Short circuit current gain
D) Short circuit input impedance
- The short-circuit admittance matrix of a two-port network is as follows. $[0; -0.5; 0.5; 0]$ Then the 2 port network is _____**
A) Non-reciprocal and passive
B) Reciprocal and passive
C) Non-reciprocal and active
D) Reciprocal and active
- The unit in which the Z-parameter are expressed is _____**
A) Mho
B) Farad
C) Ohm
D) Henry

12. The ABCD constants of this series impedance are [1, 8, 6, 3] which of the following represents transfer impedance with the output short-circuited.
 A) 1 B) 8
 C) 6 D) 3
13. What is the characteristic of a good control system?
 A) Insensitive to the parameter variation but sensitive to the input commands B) Neither sensitive to parameter variations nor sensitive to input commands
 C) Insensitive to the input command D) Sensitive to parameter variation
14. A control system working under unknown random actions is called _____.
 A) Adaptive control system B) Stochastic control system
 C) Computer control system D) Digital data system
15. The main objective of drawing root locus plot is:
 A) To obtain a clear picture about the open loop poles and zeroes of the system B) To obtain a clear picture about the transient response of feedback system for various values of open loop gain K
 C) To determine sufficient condition for the value of 'K' that will make the feedback system unstable D) Both b and c
16. Root locus is used to calculate:
 A) Relative stability B) Conditional stability
 C) Absolute stability D) Marginal stability
17. If the gain of the open loop system is doubled, the gain of the system is:
 A) One fourth of the original value B) Doubled
 C) Halved D) Not affected
18. The critical value of gain for a system is 40 and gain margin is 6dB. The system is operating at a gain of:
 A) 20 B) 40
 C) 80 D) 120
19. The roots of the characteristic equation of the second order system in which real and imaginary part represents the:
 A) Damped frequency and damping B) Damping and damped frequency
 C) Natural frequency and damping ratio D) Damping ratio and natural frequency
20. For a stable closed loop system, the gain at phase crossover frequency should always be:
 A) < 20 dB B) < 6 dB
 C) > 0 dB D) > 6 dB

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

Marks BT* CO* PO*

1. a) Determine the current i_o for the circuit shown in Figure. Q1(a) using mesh analysis.

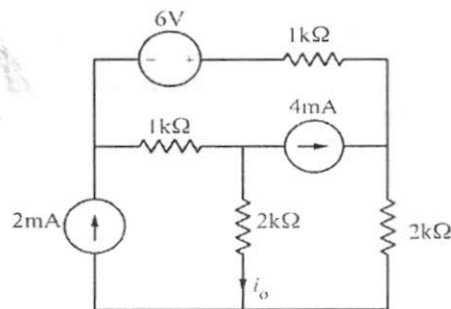


Figure. Q1(a)

8 L*3 1 1

- o) Use Norton's theorem to find current through 8Ω resistor in the circuit shown in Figure. Q1(b)

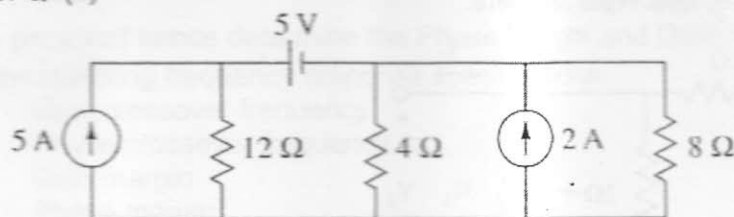


Figure. Q1(b)

8 L3 2 1

2. a) Using source transformation techniques, find i_1 for the network shown in Figure. Q2(a).

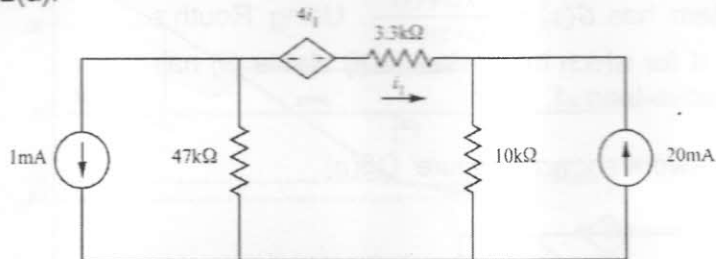


Figure. Q2(a)

8 L3 1 1

- b) Find the V_1 and V_2 using nodal analysis for the network shown in Figure. Q2(b).

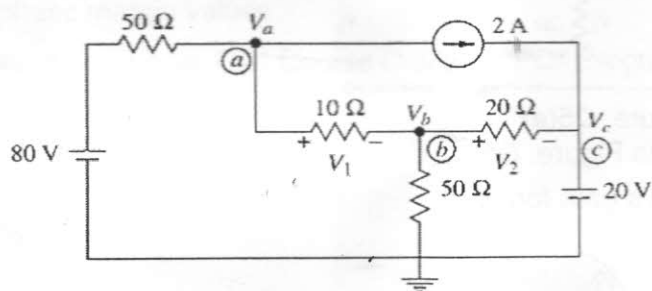


Figure. Q2(b)

8 L3 1 1

3. a) Use Superposition theorem to find voltage V_{AB} in the network shown in Figure. Q3(a)

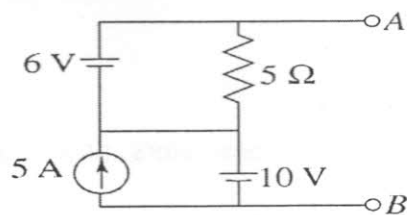


Figure. Q3(a)

8 L3 2 1

- b) Calculate the current through 24Ω resistor using Thevenin's theorem, for the circuit shown in Figure. Q3(b).

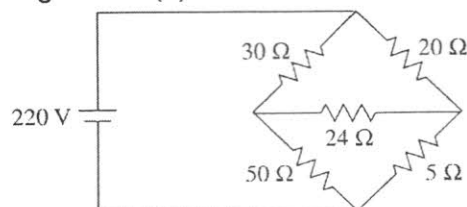


Figure. Q3(b)

8 L3 2 1

Unit – II

4. a) For the network in Figure. Q4(a), find Y parameters.

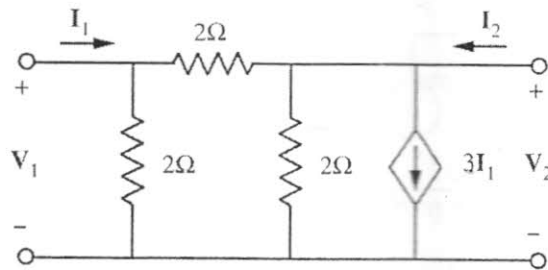


Figure. Q4(a)

8 L3 3 1

- b) A unity feedback control system has $G(s) = \frac{K(s+13)}{s(s+3)(s+7)}$. Using Routh's criterion calculate the range of K for which the system is (i) stable (ii) has its closed loop, poles more negative than -1.

8 L2 4 2

5. a) Find the Z parameters for the network shown in Figure. Q5(a).

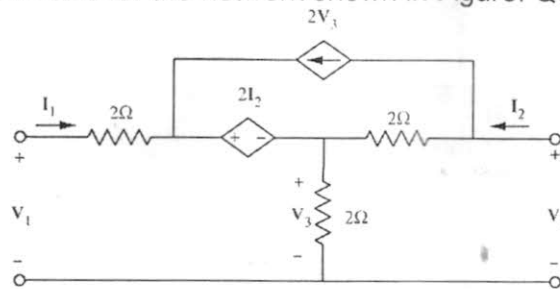


Figure. Q5(a)

8 L3 13 1

- b) For the signal flow graph shown in Figure. Q5(b), determine the transfer function $\frac{C(s)}{R(s)}$ by applying Mason's gain formula.

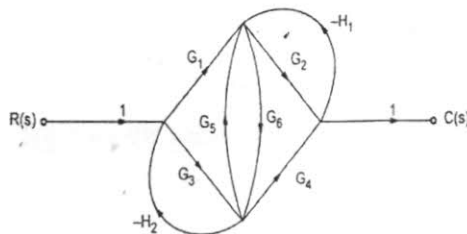


Figure. Q5(a)

8 L2 4 2

6. a) Find expressions for Z-parameters in terms of Y-parameters and H-parameters.

8 L2 3 1

- b) Determine the transfer function $\frac{E_o(s)}{E_i(s)}$ for cascaded RC circuit shown in Figure. Q6(b).

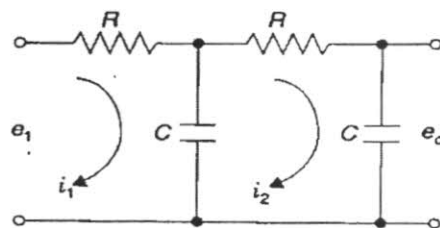


Figure. Q6(b)

8 L3 4 2

Unit – III

- a) For a unity feedback control system with $G(s) = \frac{80000}{s(s+2)(s+50)(s+200)}$, draw the bode plots and hence determine the Phase Margin and Gain Margin.

8 L3 5 1

- b) Explain the following frequency response specifications:

- Gain crossover frequency
- Phase crossover frequency
- Gain margin
- Phase margin

8 L2 5 1

8. a) Derive the transfer function from the bode plot in Figure. 8(a).

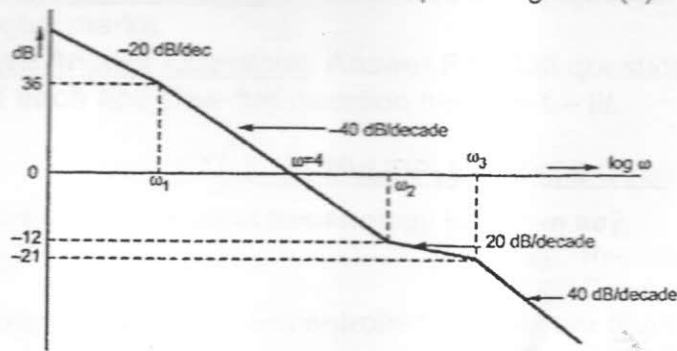


Figure. 8(a)

8 L3 5 1

- b) Explain stable, unstable and marginally stable system with respect to gain margin and phase margin values.

8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
